

FoodLens

AI-Powered Ingredient Scanner and Health Risk Scoring System

Requirement Gathering and Feasibility Analysis — MCA Capstone Project

Part A: Requirement Gathering

1. Stakeholders

- End Users (Consumers) — individuals scanning packaged products to check ingredient safety and overall health impact, including those managing chronic health conditions or specific food allergies.
- Community Contributors — users who submit ingredient data for regional or local products that are missing from existing global product databases, helping close the regional coverage gap over time.
- Project Team (4 Members) — responsible for the design, development, testing, and deployment of the system across its input, processing, scoring, and presentation modules.
- Project Guide/Evaluator — the academic supervisor responsible for assessing the project's scope, technical feasibility, and overall execution.
- Administrator — reviews and approves community-submitted product data, and monitors system-wide analytics for data quality.

2. Functional Requirements

- The system shall allow users to scan a product barcode/QR code using the device camera, and capture a photograph of the ingredient label directly when no barcode is available or recognized.
- The system shall query the Open Food Facts API to retrieve product and ingredient data for recognized barcodes.
- The system shall perform OCR on label images when no barcode match is found, and display an OCR confidence score to the user.
- The system shall clean, normalize, and resolve ingredient aliases (e.g., mapping the additive code E211 to Sodium Benzoate) using NLP techniques.
- The system shall compute a deterministic health risk score using the rule-based ingredient database, weighting risk by ingredient category and position in the ingredient list, and shall provide an expandable, ingredient-level breakdown explaining the computed score.
- The system shall allow users to create a health profile specifying chronic conditions (e.g., diabetes, hypertension) and allergies, and shall adjust the health score using condition-specific multipliers based on this profile.
- The system shall issue a hard allergen warning independent of the numeric score whenever an allergen is detected, and shall support multiple health profiles under a single shared family account.
- The system shall generate a plain-language explanation of the score using a large language model constrained to pre-verified data.
- The system shall support dark mode and general accessibility settings.
- The system shall allow users to compare two scanned products side by side and receive healthier-alternative and recipe suggestions.
- The system shall track a user's consumption history, display daily/weekly trends, and generate exportable PDF reports and scheduled email health reports.
- The system shall allow users to submit ingredient data for unrecognized regional products, routed through admin review or multi-user consensus approval, and shall provide an admin dashboard with aggregate analytics.
- The system shall generate a shareable QR code or link for a scanned product's results.

3. Non-Functional Requirements

- Performance — barcode lookup results should be returned within 2-3 seconds under normal network conditions, and OCR processing should complete within 5-8 seconds per image.

- Reliability — the scoring engine must produce consistent, deterministic results for the same ingredient list and user profile every time.
- Security & Privacy — user health profile data, including conditions and allergies, must be stored securely and never exposed to other users.
- Data Accuracy — every health-related claim shown to the user must be traceable to a cited authoritative source such as FSSAI or WHO/ICMR.
- Scalability & Maintainability — the ingredient database and community submission system should support growth in both users and product entries, with modules independently testable and deployable to support parallel four-member development.

4. Data Requirements

- Ingredient Risk Database — ingredient name, category, base risk score, aliases, allergen flag, source citation, and condition multipliers.
- Product Database — product name, brand, barcode, ingredient list, and data source (Open Food Facts or community-verified).
- User Profile Data — user ID, health conditions, allergies, and linked family profiles.
- Scan History Data — user ID, product ID, timestamp, and computed score.
- Community Submission Data — product details, submitted ingredient list, submission status, submitter ID, and consensus count.

5. Constraints and Assumptions

- The system is dependent on third-party APIs (Open Food Facts, OCR service, LLM provider) for core functionality, and OCR accuracy may vary on poor-quality or damaged product labels.
- The initial ingredient risk database is limited to approximately 100-150 common ingredients due to project timeline constraints, and the system must be developed and demonstrated within a single academic semester by a four-member team.
- It is assumed that users have a smartphone with a working camera and internet connectivity, that self-reported health condition and allergy data is accurate, and that the Open Food Facts API remains stable and available during development and demonstration.

Part B: Feasibility Analysis

1. Technical Feasibility

The project relies on mature, well-documented technologies with low implementation risk. Barcode/product lookup uses the Open Food Facts API, a free, open, and stable public data source requiring no special licensing. OCR is achievable using mature open-source (Tesseract) or cloud-based (Google Cloud Vision) services with well-documented APIs, and NLP-based ingredient parsing and alias resolution can be implemented using standard libraries such as spaCy and regex, without requiring custom model training. The scoring engine is a deterministic, rule-based system built on standard backend logic rather than complex machine learning, which keeps implementation risk low and results easy to verify. LLM integration is scoped narrowly to OCR text cleanup and explanation phrasing, both low-complexity, well-supported use cases. The main technical risks — OCR accuracy on real-world regional packaging and maintaining consistent alias-matching coverage — are manageable through iterative testing on a representative product sample rather than requiring novel research.

2. Operational Feasibility

The system is intended for ordinary smartphone users with no specialized training, and its core workflow of scanning a product, viewing a score, and reading an explanation mirrors patterns already familiar from existing consumer apps, minimizing adoption friction. Community contribution and admin review workflows are simple, form-based interactions requiring no specialized training for contributors or the reviewing administrator. The four-member team structure maps

cleanly onto four independent functional areas — input/lookup, OCR/NLP, scoring/personalization, and explanation/UI — supporting realistic parallel development without excessive cross-dependency during the build phase.

3. Economic Feasibility

The project can be developed and demonstrated at effectively zero to minimal cost, which is appropriate for an academic project with no external funding. The Open Food Facts API, Tesseract OCR, MySQL, and the React/React Native and Flask/Django frameworks are all free and open-source. Google Cloud Vision API, if used instead of Tesseract, offers a free tier sufficient for development and demonstration. LLM API usage is limited to narrowly scoped calls for OCR cleanup and explanation generation, keeping token usage low enough to fit within free-tier or minimal-cost allowances. No specialized hardware is required beyond standard development laptops and a smartphone for testing, meaning the project requires no meaningful capital investment.

4. Legal and Data Feasibility

Product and ingredient data sourced from Open Food Facts is published under an open license permitting reuse for non-commercial and academic purposes, and ingredient risk scoring is based on publicly available regulatory and health guidelines from FSSAI and WHO/ICMR, which are freely citable for academic and informational use. The system explicitly avoids presenting AI-generated content as an unverified medical diagnosis, framing all output as informational scoring rather than medical advice, which reduces liability concerns appropriate for a non-commercial academic prototype. User health profile data should be handled with basic data protection practices, such as not exposing one user's health data to another, even at prototype scale.

5. Overall Assessment

FoodLens is technically, operationally, economically, and legally feasible to build and demonstrate as a four-member MCA major project within a single academic semester. The primary risk areas — OCR accuracy on real-world regional packaging and the breadth of the initial ingredient risk database — are manageable through a deliberately scoped dataset of approximately 100-150 ingredients and iterative testing, rather than requiring novel research or significant additional investment. The modular architecture aligns well with a four-person team, allowing parallel development with clearly defined integration points, making the overall project plan realistic and low-risk.